Study the lesson below. Reflect on ways in which this teacher helped students to construct their own knowledge.

CONSTRUCTING KNOWLEDGE

This page is out of my lesson note book. Its purpose is to remind me of the content for this lesson. Can you as experienced teachers construct my lesson from these notes? Could a novice teacher presented with this plan implement this lesson? Are the numbers presented random or put together with purpose?

How do I put together lessons, how do I work in my classroom?

I will attempt to give an account of how this lesson went. They are a year 9 group who has been doing some work previously on multiplying and dividing with powers of ten. This lesson also had two novice teachers observing for the first time. The lesson started with me asking pupils to come and get a school scientific calculator (we use casio f x 82). Switch on the calculators and press [mode] [9].

Input the number 2.346 and multiply it by 10, keep multiplying the answer by ten until you notice something strange happening. The pupils record each answer as they went. Pupils stopped when the display showed 2.346 ^{08}

How many tens had you multiplied by? So what did this display mean? Pupils were happy to show this as

2.346×10^8

I then asked them to input 500. then to press the buttons [mode] [8] [8]. What did the calculator show? How would you write this in standard form?

The pupils were then told that I was going to give them ten numbers (please refer to lesson notes sheet) that they had to do this process on and they were to record in their books the input number, the calculator display and the standard form representation in this table.

number	calculator display	standard form

They were then told that they would be given another ten numbers which they would have to put into standard form but without a calculator. So they had to focus on the first and the last columns and could they spot a quick method of doing this for when we did not have a calculator.

The pupils settled to the task. I circulated around the room to remind individuals to press [mode] [8] [8], to check that pupils were achieving correct answers and to focus pupils on how to go from the first to the last column without the calculator.

When pupils began to finish I asked them to write down their rule for doing this, to pick 5 numbers of their own, use their rule and then to check, using the calculator, when all the pupils

had completed the table we went around the room filling in the answers on the board. Asking if anyone had got a different answer. This threw up answers such as

4.8000000×10^{03} and 4.8×10^{3}

as being different. I at this point asked pupils to identify what was the same about the two answers and what was different about them. This identified the noughts as being not significant and pupils came to view these as identical answers. I then asked if every pupil had a method for going from the first column to the last without a calculator which they had. We then closed the calculators and in silence I put up the second set of numbers. (see lesson note sheets). At this stage I had not asked the pupils to share their methods with others. We then marked the pupils answers to the set by me stating the answers. I then asked all the people with ten out of ten to put up their hands, about eight. How many had nine out of ten? The rest of the class. I asked which one had caused the problem. Number 7 (were the questions a random choice?) A pupil who had got ten right explained how they did it. The class was asked to focus on what they needed to change to their rule in the light of this question. Pupils recognized what they had to do. When asked if they would forget this method they replied no. on the way out Claire said its just like riding a bike! (She had been taught by me last year.)

The important aspects of this lesson plan for me are pupils having access to a pool of information created in this case by the calculator display on which they can draw. The pupils being focused on not just pressing calculator keys but on the connection between the number and the standard form representation. This focus was honed by stating that for the second set you would not have a calculator. The choice of the ten numbers, to include the first three which had a supportive feel to them, the challenge question at number seven. This forced pupils to move from justifying from the right to focusing on the decimal point which caused them to alter their methods. It is important to note that this group could have given me correct answers based on an incomplete knowledge of standard form. The fact that I had not told them anything other than instructions. That they had recorded their own rules in their own words.

I would welcome writings of this sort focusing on just a good idea. To focus on how we actually work, what is the art of teaching?

This was the final exercise.
Write in standard form.

1. 9000
2. 90 000
3. 2700
4. 493 200
5. 0.78
6. 0.078
7. 29.84
8. 0.00786
9. 0.003042
10. 0.1010101